

BSc (Hons) Computer Science
BSc (Hons) Cyber Security
Module Code: QH0305 Module Title: Problem Solving
Assessment Sheet 3

Instructions:

This is one of the eight assessment tasks which will contribute to the overall mark. You will need to complete the tasks as outlined below and then document them in a Word document file. As a minimum, you should provide screenshots of the following:

- Your code
- The output that your code generates

In instances where your code could generate different outputs depending on what values are given, you should provide multiple screenshots of the console screen, showing the different outputs in order to demonstrate that the code works correctly.

Your Word document should have appropriate headings to ensure that each task can easily be identified alongside the rest of your work.

This assessment will focus on conversions and use of loops.

You must attempt all tasks on this sheet to achieve a higher grade. For example, if you want to gain marks between 70 - 100, you must complete all other grades first and add them to your portfolio with screenshots.

A zip folder with all Grade codes (in .c or .txt format) must be submitted alongside the portfolio (MS Word file). This is in addition to the code presented as screenshots.

Assessment Task 3: Hydration Tracker Loop Challenge

Create a program that tracks **daily water intake** over time. The user will input data as **millilitres**, **bottles**, or **cups**. The program must use **loops** and **conditional statements** to manage functionality and produce summaries.

To achieve Grade D (Between 40-49) (Basic Conversion)

Requirements:

1. Write a program that converts a single day's water intake (in millilitres) into litres and cups.
2. Prompt the user to input the number of millilitres consumed in one day.
3. Use fixed conversion rates:
 - 1 litre = 1000 ml
 - 1 cup = 240 ml
4. Display the total litres and equivalent cups (round cups to 2 decimal places).

Example Interaction:

Enter water intake for today (ml): 1800

Litres: 1.80 L

Cups (240 ml): 7.50 cups

To achieve Grade C (Between 50-59) (Multi-day Tracking)
Requirements:

1. Complete all previous steps.
2. Extend the program to allow the user to input data for multiple days.
3. Use a loop to repeatedly prompt for a day's intake in millilitres and perform the conversions.
4. After each entry, ask the user if they want to enter another day.
5. When the user quits, display the number of days tracked and the total litres consumed across those days.

Example Interaction:

Enter water intake for today (ml): 1500

Litres: 1.50 L | Cups: 6.25

Add another day? (y/n): y

Enter water intake for today (ml): 2200

Litres: 2.20 L | Cups: 9.17

Add another day? (y/n): n

You tracked 2 days. Total: 3.70 L

To achieve Grade B (Between 60-69) (Multi-Type Tracking)
Requirements:

1. Complete all previous steps.
2. Add a menu that lets the user choose what to add to the record for the current day:
 - 1: Millilitres (ml)
 - 2: Bottles (assume 1 bottle = 500 ml)
 - 3: Cups (assume 1 cup = 240 ml)
3. Convert any selected unit to millilitres, then update running totals.
4. Allow the user to return to the menu or exit after each entry.
5. On exit, display a summary showing: total ml, total litres, and how many entries of each type were recorded.

Example Interaction:

What would you like to add?

1: Millilitres

2: Bottles (500 ml each)

3: Cups (240 ml each)

Enter your choice: 2

How many bottles? 3

Added 1500 ml.

Add another entry? (y/n): y

What would you like to add?

1: Millilitres

2: Bottles

3: Cups

Enter your choice: 3

How many cups? 4

Added 960 ml.

Add another entry? (y/n): n

Summary:

Total: 2460 ml (2.46 L)

Entries — Bottles: 1, Cups: 1, Millilitres: 0

To achieve Grade A (Between 70-100)**Requirements:**

Complete all previous steps, then add:

- A daily hydration goal set when the program starts (in litres). After each entry, display progress as a percentage of the goal (based on today's total).
- An environment factor that affects suggested intake when adding cups or bottles:
 - Cool/Indoor: no adjustment
 - Warm/Active: +10% suggested daily target
 - Hot/Very Active: +20% suggested daily target
(This factor should affect the daily target.)
- Robust input validation for all user inputs, including:
 - Numeric values must be positive
 - Handle invalid menu choices gracefully
 - Re-prompt until a valid entry is received
- On exit, show:
 - Per-day summary (total ml and litres per day)
 - Overall totals across all days and days meeting/exceeding the daily goal
 - The longest streak (in days) of meeting/exceeding the goal

Assignment Preparation Guidelines

- All components of the assignment report must be Word-processed (**handwritten text or hand drawn diagrams are not acceptable**), font size must be within the range of 11 point to 14 point including the headings, body text and any texts within diagrams.
- Standard and commonly used fonts such as Times New Roman, Arial or Calibri should be used.
- All figures, graphs and tables must be numbered and labelled with short explanations.
- Material from external sources must be properly acknowledged and cited within the text using the Harvard referencing system.
- All components of the assignment (text, diagrams, code etc.) and all assessment sheets must be submitted in one Word file.
- The report should be logically structured, the core of the report may start by defining the problem / requirements, followed by the proposed solution including a detailed discussion, analysis and evaluation, leading to the implementation and testing stage, finally a conclusion and personal reflection on learning.
- Screenshots without description / discussion are not suitable as they do not express your understanding or support your work adequately.

Submission instructions

- This is a portfolio assignment with eight tasks in total. Each task will be completed and saved in the portfolio. Once the portfolio is completed, it should be submitted on Turnitin. The submission link to Turnitin can be found under the “Assessment Tab” in your module section in the SOL VLE.
- Please note file size limitation might apply. Your report must be under 250MB.
- The source code (in .c or .txt format) for each task should be **zipped** and complete single ZIP file **submitted** to the VLE alongside the report.
- The Assignment Brief can be found under the “Assessment Tab” in your module section in the SOL VLE.
- **Refer to the Assignment Brief** to find the links to Late Submissions, Extenuating Circumstances, Academic Misconduct, Ethics Policy, Grade marking and Guidance for online submission through Solent Online Learning (SOL).